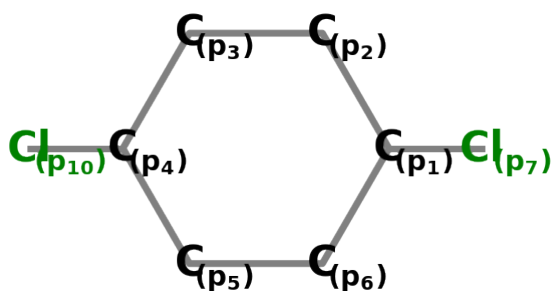


Dichlorobenzene



Point Group: **D_{2h}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 8E ; -4C_2(y) ; -8\sigma(xy) ; 4\sigma(yz)$$

$$\Gamma_{irred} = 1B_{2g} \oplus 3B_{3g} \oplus 1A_u \oplus 3B_{1u}$$

- SALC of B_{3g}: $\frac{\sqrt{2}(p_1 - p_4)}{2}$
- SALC of B_{1u}: $\frac{\sqrt{2}(p_1 + p_4)}{2}$
- SALC of B_{2g}: $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$
- SALC of B_{3g}: $\frac{p_2}{2} - \frac{p_3}{2} - \frac{p_5}{2} + \frac{p_6}{2}$
- SALC of A_u: $\frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2}$
- SALC of B_{1u}: $\frac{p_2}{2} + \frac{p_3}{2} + \frac{p_5}{2} + \frac{p_6}{2}$
- SALC of B_{3g}: $\frac{\sqrt{2}(p_7 - p_8)}{2}$
- SALC of B_{1u}: $\frac{\sqrt{2}(p_7 + p_8)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B_{3g}} = \begin{pmatrix} \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 0 \\ \sqrt{2}\tilde{\beta} & \tilde{\alpha} - \tilde{\beta} & 0 \\ 0 & 0 & \tilde{\alpha}_{Cl} \end{pmatrix}$$

$$H_{B_{1u}} = \begin{pmatrix} \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 2\tilde{\beta}_{Cl} \\ \sqrt{2}\tilde{\beta} & \tilde{\alpha} + \tilde{\beta} & 0 \\ 2\tilde{\beta}_{Cl} & 0 & \tilde{\alpha}_{Cl} \end{pmatrix}$$

$$H_{B_{2g}} = (\tilde{\alpha} + \tilde{\beta})$$

$$H_{A_u} = (\tilde{\alpha} - \tilde{\beta})$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of B_{1u} with energy = $\frac{\left(4\tilde{\alpha} + 2\tilde{\alpha}_{Cl} + 2\tilde{\beta} - 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} + \sqrt{-4(-3\tilde{\alpha}^2 - 6\tilde{\alpha}\tilde{\alpha}_{Cl} - 3\tilde{\alpha}\tilde{\beta} - 3\tilde{\alpha}_{Cl}\tilde{\beta} + 3\tilde{\alpha}_{Cl}\tilde{\beta}_{Cl} + 3\tilde{\beta}\tilde{\beta}_{Cl})}}{2}\right)}{2}$

(

$$(\text{?}_1 \text{Norm}^2 \quad \text{?}_2 \text{Norm}^2 \quad \text{?}_3 \text{Norm}^2) * \begin{pmatrix} \text{1. SALC in B1u} \\ \text{2. SALC in B1u} \\ \text{3. SALC in B1u} \end{pmatrix} = (\text{?}_1 \text{Norm}^2 \quad \text{?}_2 \text{Norm}^2 \quad \text{?}_3 \text{Norm}^2) * \begin{pmatrix} \frac{p_2}{2} \\ \frac{p_3}{2} \\ \frac{p_5}{2} \end{pmatrix}$$

$$= \text{?}_1 \text{Norm}^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right) + \text{?}_2 \text{Norm}^2 \left(\frac{p_2}{2} + \frac{p_3}{2} + \frac{p_5}{2} + \frac{p_6}{2} \right) + \text{?}_3 \text{Norm}^2 \left(\frac{\sqrt{2}p_7}{2} + \frac{\sqrt{2}p_8}{2} \right)$$

)

• orbital of B1u with energy = $\frac{(1+\sqrt{3}i) \left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}(1+\sqrt{3}i) \right)^3 \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}}}{(1+\sqrt{3}i) \left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}(1+\sqrt{3}i) \right)^3 \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}}}$

(

$$(\text{?}_1 Norm^2 \quad \text{?}_2 Norm^2 \quad \text{?}_3 Norm^2) * \begin{pmatrix} 1. \text{ SALC in B1u} \\ 2. \text{ SALC in B1u} \\ 3. \text{ SALC in B1u} \end{pmatrix} = (\text{?}_1 Norm^2 \quad \text{?}_2 Norm^2 \quad \text{?}_3 Norm^2) * \begin{pmatrix} \frac{p_2}{2} \end{pmatrix}$$

$$= \text{?}_1 Norm^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right) + \text{?}_2 Norm^2 \left(\frac{p_2}{2} + \frac{p_3}{2} + \frac{p_5}{2} + \frac{p_6}{2} \right) + \text{?}_3 Norm^2 \left(\frac{\sqrt{2}p_7}{2} + \frac{\sqrt{2}p_8}{2} \right)$$

)

• orbital of B1u with energy = $\frac{(1-\sqrt{3}i) \left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}(1-\sqrt{3}i) \right)^3 \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}}}{(1-\sqrt{3}i) \left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}(1-\sqrt{3}i) \right)^3 \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}}}$

(

$$(\text{?}_1 Norm^2 \quad \text{?}_2 Norm^2 \quad \text{?}_3 Norm^2) * \begin{pmatrix} 1. \text{ SALC in B1u} \\ 2. \text{ SALC in B1u} \\ 3. \text{ SALC in B1u} \end{pmatrix} = (\text{?}_1 Norm^2 \quad \text{?}_2 Norm^2 \quad \text{?}_3 Norm^2) * \begin{pmatrix} \frac{p_2}{2} \end{pmatrix}$$

$$= \text{?}_1 Norm^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right) + \text{?}_2 Norm^2 \left(\frac{p_2}{2} + \frac{p_3}{2} + \frac{p_5}{2} + \frac{p_6}{2} \right) + \text{?}_3 Norm^2 \left(\frac{\sqrt{2}p_7}{2} + \frac{\sqrt{2}p_8}{2} \right)$$

)

• orbital of B3g with energy = $\tilde{\alpha} + \tilde{\beta}$

(

$$\begin{pmatrix} \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B3g} \\ 2. \text{ SALC in B3g} \\ 3. \text{ SALC in B3g} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}p_1}{2} - \frac{\sqrt{2}p_4}{2} \\ \frac{p_2}{2} - \frac{p_3}{2} - \frac{p_5}{2} + \frac{p_6}{2} \\ \frac{\sqrt{2}p_7}{2} - \frac{\sqrt{2}p_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{3}(2p_1 + p_2 - p_3 - 2p_4 - p_5 + p_6)}{6}$$

)

• orbital of B3g with energy = $\tilde{\alpha}_{Cl}$

(

$$\begin{pmatrix} 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B3g} \\ 2. \text{ SALC in B3g} \\ 3. \text{ SALC in B3g} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}p_1}{2} - \frac{\sqrt{2}p_4}{2} \\ \frac{p_2}{2} - \frac{p_3}{2} - \frac{p_5}{2} + \frac{p_6}{2} \\ \frac{\sqrt{2}p_7}{2} - \frac{\sqrt{2}p_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{2}(p_7 - p_8)}{2}$$

)

- orbital of B3g with energy = $\tilde{\alpha} - 2\tilde{\beta}$
(

$$\begin{pmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B3g} \\ \text{2. SALC in B3g} \\ \text{3. SALC in B3g} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}p_1}{2} - \frac{\sqrt{2}p_4}{2} \\ \frac{p_2}{2} - \frac{p_3}{2} - \frac{p_5}{2} + \frac{p_6}{2} \\ \frac{\sqrt{2}p_7}{2} - \frac{\sqrt{2}p_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-p_1 + p_2 - p_3 + p_4 - p_5 + p_6)}{6}$$

)

- orbital of B2g with energy = $\tilde{\alpha} + \tilde{\beta}$
(

$$\begin{pmatrix} 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B2g} \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} * \left(\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2} \right)$$

$$= \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$$

)

- orbital of Au with energy = $\tilde{\alpha} - \tilde{\beta}$
(

$$\begin{pmatrix} 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in Au} \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} * \left(\frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2} \right)$$

$$= \frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2}$$

)

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 8E ; 4C_2(y) ; 8\sigma(xy) ; 4\sigma(yz)$$

$$\Gamma_{irred} = 3A_g \oplus 1B_{1g} \oplus 3B_{2u} \oplus 1B_{3u}$$

- SALC of A_g : $\frac{\sqrt{2}(s_1+s_4)}{2}$
- SALC of B_{2u} : $\frac{\sqrt{2}(s_1-s_4)}{2}$
- SALC of A_g : $\frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2}$
- SALC of B_{1g} : $\frac{s_2}{2} - \frac{s_3}{2} + \frac{s_5}{2} - \frac{s_6}{2}$
- SALC of B_{2u} : $\frac{s_2}{2} - \frac{s_3}{2} - \frac{s_5}{2} + \frac{s_6}{2}$
- SALC of B_{3u} : $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$
- SALC of A_g : $\frac{\sqrt{2}(s_7+s_8)}{2}$
- SALC of B_{2u} : $\frac{\sqrt{2}(s_7-s_8)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{A_g} = \begin{pmatrix} \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 2\beta_{s_{Cl}} \\ \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s + \tilde{\beta}_s & 0 \\ 2\beta_{s_{Cl}} & 0 & \alpha_{s_{Cl}} \end{pmatrix}$$

$$H_{B_{2u}} = \begin{pmatrix} \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 0 \\ \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s - \tilde{\beta}_s & 0 \\ 0 & 0 & \alpha_{s_{Cl}} \end{pmatrix}$$

$$H_{B_{1g}} = (\tilde{\alpha}_s - \tilde{\beta}_s)$$

$$H_{B_{3u}} = (\tilde{\alpha}_s + \tilde{\beta}_s)$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A_g with energy = $\frac{4\tilde{\alpha}_s + 2\alpha_{s_{Cl}} + 2\tilde{\beta}_s - 2^{2/3} \sqrt[3]{-27\tilde{\alpha}_s^2\alpha_{s_{Cl}} - 27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s + 108\tilde{\alpha}_s\beta_{s_{Cl}}^2 + 54\alpha_{s_{Cl}}\tilde{\beta}_s^2 + 108\beta_{s_{Cl}}^3}}}{3}$

(

$$(\text{?}_1 \text{Norm}^2 \quad \text{?}_2 \text{Norm}^2 \quad \text{?}_3 \text{Norm}^2) * \begin{pmatrix} 1. \text{ SALC in } A_g \\ 2. \text{ SALC in } A_g \\ 3. \text{ SALC in } A_g \end{pmatrix} = (\text{?}_1 \text{Norm}^2 \quad \text{?}_2 \text{Norm}^2 \quad \text{?}_3 \text{Norm}^2) * \begin{pmatrix} \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \\ \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \\ \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$= \text{?}_1 \text{Norm}^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right) + \text{?}_2 \text{Norm}^2 \left(\frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \right) + \text{?}_3 \text{Norm}^2 \left(\frac{\sqrt{2}s_7}{2} + \frac{\sqrt{2}s_8}{2} \right)$$

)

• orbital of Ag with energy =
$$\frac{(1+\sqrt{3}i) \left(8\tilde{\alpha}_s + 4\alpha_{s_{Cl}} + 4\tilde{\beta}_s + 2^{2/3}(1+\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}_s^2\alpha_{s_{Cl}} - 27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s + 108\tilde{\alpha}_s\beta_{s_{Cl}}^2} \right)}{}$$

(

$$(\begin{matrix} ?_1 Norm^2 & ?_2 Norm^2 & ?_3 Norm^2 \end{matrix}) * \begin{pmatrix} 1. \text{ SALC in Ag} \\ 2. \text{ SALC in Ag} \\ 3. \text{ SALC in Ag} \end{pmatrix} = (\begin{matrix} ?_1 Norm^2 & ?_2 Norm^2 & ?_3 Norm^2 \end{matrix}) * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \\ \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \\ \frac{\sqrt{2}s_7}{2} + \frac{\sqrt{2}s_8}{2} \end{pmatrix}$$

$$=?_1 Norm^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right) + ?_2 Norm^2 \left(\frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}s_7}{2} + \frac{\sqrt{2}s_8}{2} \right)$$

)

• orbital of Ag with energy =
$$\frac{(1-\sqrt{3}i) \left(8\tilde{\alpha}_s + 4\alpha_{s_{Cl}} + 4\tilde{\beta}_s + 2^{2/3}(1-\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}_s^2\alpha_{s_{Cl}} - 27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s + 108\tilde{\alpha}_s\beta_{s_{Cl}}^2} \right)}{}$$

(

$$(\begin{matrix} ?_1 Norm^2 & ?_2 Norm^2 & ?_3 Norm^2 \end{matrix}) * \begin{pmatrix} 1. \text{ SALC in Ag} \\ 2. \text{ SALC in Ag} \\ 3. \text{ SALC in Ag} \end{pmatrix} = (\begin{matrix} ?_1 Norm^2 & ?_2 Norm^2 & ?_3 Norm^2 \end{matrix}) * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \\ \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \\ \frac{\sqrt{2}s_7}{2} + \frac{\sqrt{2}s_8}{2} \end{pmatrix}$$

$$=?_1 Norm^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right) + ?_2 Norm^2 \left(\frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}s_7}{2} + \frac{\sqrt{2}s_8}{2} \right)$$

)

• orbital of B2u with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$

(

$$\begin{pmatrix} \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B2u} \\ 2. \text{ SALC in B2u} \\ 3. \text{ SALC in B2u} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} - \frac{\sqrt{2}s_4}{2} \\ \frac{s_2}{2} - \frac{s_3}{2} - \frac{s_5}{2} + \frac{s_6}{2} \\ \frac{\sqrt{2}s_7}{2} - \frac{\sqrt{2}s_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{3}(2s_1 + s_2 - s_3 - 2s_4 - s_5 + s_6)}{6}$$

)

• orbital of B2u with energy = $\alpha_{s_{Cl}}$

(

$$\begin{pmatrix} 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B2u} \\ 2. \text{ SALC in B2u} \\ 3. \text{ SALC in B2u} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} - \frac{\sqrt{2}s_4}{2} \\ \frac{s_2}{2} - \frac{s_3}{2} - \frac{s_5}{2} + \frac{s_6}{2} \\ \frac{\sqrt{2}s_7}{2} - \frac{\sqrt{2}s_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{2}(s_7 - s_8)}{2}$$

)

- orbital of B2u with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$

(

$$\begin{pmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B2u} \\ \text{2. SALC in B2u} \\ \text{3. SALC in B2u} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}\mathbf{s}_1}{2} - \frac{\sqrt{2}\mathbf{s}_4}{2} \\ \frac{\mathbf{s}_2}{2} - \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} + \frac{\mathbf{s}_6}{2} \\ \frac{\sqrt{2}\mathbf{s}_7}{2} - \frac{\sqrt{2}\mathbf{s}_8}{2} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3 + \mathbf{s}_4 - \mathbf{s}_5 + \mathbf{s}_6)}{6}$$

)

- orbital of B3u with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$

(

$$\begin{pmatrix} 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B3u} \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} * \left(\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2} \right)$$

$$= \frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$$

)

- orbital of B1g with energy = $\tilde{\alpha}_s - \tilde{\beta}_s$

(

$$\begin{pmatrix} 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B1g} \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} * \left(\frac{\mathbf{s}_2}{2} - \frac{\mathbf{s}_3}{2} + \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2} \right)$$

$$= \frac{\mathbf{s}_2}{2} - \frac{\mathbf{s}_3}{2} + \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$$

)

Transitions from p orbitals into s orbitals

State with Occupation (2, 2, 2, 0, 0, 0, 2, 2) (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 2, 0, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i) \left(52\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 8\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}$$

- Energy of State after Excitation:
$$\frac{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 2, 0, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i) \left(52\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 8\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}$$

- Energy of State after Excitation:
$$\frac{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s + 2\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0, 2, 2) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 1, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 1, 1, 1, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}-\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}$$

- Energy of State after Excitation:
$$\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 1, 1, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 0, 1, 1, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}-\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}$$

- Energy of State after Excitation:
$$\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-\right)}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 1, 1, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2, 2), (0, 0, 0, 1, 0, 0, 0, 0)$

- ψ_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right)$, energy = $\frac{(1+\sqrt{3}i) \left(8\tilde{\alpha} + 4\tilde{\alpha}_{Cl} + 4\tilde{\beta} + 2^{2/3} (1+\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}^2 \tilde{\alpha}_{Cl}} \right)}{3 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right)}$

- η_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right)$, energy = $\frac{(1+\sqrt{3}i) \left(8\tilde{\alpha}_s + 4\alpha_{s_{Cl}} + 4\tilde{\beta}_s + 2^{2/3} (1+\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}_s^2 \tilde{\alpha}_{Cl}} \right)}{3 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right)}$

- Δ Energy: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{\tilde{\beta}}{3} + \frac{\tilde{\beta}_s}{3} + \frac{3 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) \sqrt[3]{-\frac{27\tilde{\alpha}^2 \tilde{\alpha}_{Cl}}{2} - \frac{27\tilde{\alpha} \tilde{\alpha}_{Cl} \tilde{\beta}}{2} + 54\tilde{\alpha} \tilde{\beta}_{Cl}^2 + 27\tilde{\alpha}_{Cl} \tilde{\beta}^2 + 54\tilde{\beta} \tilde{\beta}_{Cl}^2 + \sqrt{-4(-3\tilde{\alpha}^2 - 6\tilde{\alpha} \tilde{\alpha}_{Cl} - 3\tilde{\alpha} \tilde{\beta} - 3\tilde{\alpha}_{Cl} \tilde{\beta} + 6\tilde{\beta}^2 + 12\tilde{\beta}_{Cl}^2 + (-2\tilde{\alpha} - \tilde{\alpha}_{Cl} - \tilde{\beta})^2)}}}{3 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right)}$

$$3 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) \sqrt[3]{-\frac{27\tilde{\alpha}_s^2 \alpha_{s_{Cl}}}{2} - \frac{27\tilde{\alpha}_s \alpha_{s_{Cl}} \tilde{\beta}_s}{2} + 54\tilde{\alpha}_s \beta_{s_{Cl}}^2 + 27\alpha_{s_{Cl}} \tilde{\beta}_s^2 + 54\tilde{\beta}_s \beta_{s_{Cl}}^2 + \sqrt{-4(-3\tilde{\alpha}_s^2 - 6\tilde{\alpha}_s \alpha_{s_{Cl}} - 3\tilde{\alpha}_s \tilde{\beta}_s - 3\alpha_{s_{Cl}} \tilde{\beta}_s + 6\tilde{\beta}_s^2 + 12\beta_{s_{Cl}}^2 + (-2\tilde{\alpha}_s - \alpha_{s_{Cl}} - \tilde{\beta}_s)^2)}}}$$

$$\left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) \sqrt[3]{-\frac{27\tilde{\alpha}^2 \tilde{\alpha}_{Cl}}{2} - \frac{27\tilde{\alpha} \tilde{\alpha}_{Cl} \tilde{\beta}}{2} + 54\tilde{\alpha} \tilde{\beta}_{Cl}^2 + 27\tilde{\alpha}_{Cl} \tilde{\beta}^2 + 54\tilde{\beta} \tilde{\beta}_{Cl}^2 + \sqrt{-4(-3\tilde{\alpha}^2 - 6\tilde{\alpha} \tilde{\alpha}_{Cl} - 3\tilde{\alpha} \tilde{\beta} - 3\tilde{\alpha}_{Cl} \tilde{\beta} + 6\tilde{\beta}^2 + 12\tilde{\beta}_{Cl}^2 + (-2\tilde{\alpha} - \tilde{\alpha}_{Cl} - \tilde{\beta})^2)}}}$$

$$\left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) \sqrt[3]{-\frac{27\tilde{\alpha}_s^2 \alpha_{s_{Cl}}}{2} - \frac{27\tilde{\alpha}_s \alpha_{s_{Cl}} \tilde{\beta}_s}{2} + 54\tilde{\alpha}_s \beta_{s_{Cl}}^2 + 27\alpha_{s_{Cl}} \tilde{\beta}_s^2 + 54\tilde{\beta}_s \beta_{s_{Cl}}^2 + \sqrt{-4(-3\tilde{\alpha}_s^2 - 6\tilde{\alpha}_s \alpha_{s_{Cl}} - 3\tilde{\alpha}_s \tilde{\beta}_s - 3\alpha_{s_{Cl}} \tilde{\beta}_s + 6\tilde{\beta}_s^2 + 12\beta_{s_{Cl}}^2 + (-2\tilde{\alpha}_s - \alpha_{s_{Cl}} - \tilde{\beta}_s)^2)}}}$$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i) \left(88\tilde{\alpha} + 32\tilde{\alpha}_{Cl} - 4\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2 \tilde{\alpha}_{Cl} - 27\tilde{\alpha} \tilde{\alpha}_{Cl} \tilde{\beta} + 108\tilde{\alpha} \tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl} \tilde{\beta}^2} \right)}{(1-\sqrt{3}i)(1+\sqrt{3}i)}$

- Energy of State after Excitation: $\frac{8 \sqrt[3]{2} (1-\sqrt{3}i)(1+\sqrt{3}i) \left(3\tilde{\alpha}^2 + 6\tilde{\alpha} \tilde{\alpha}_{Cl} + 3\tilde{\alpha} \tilde{\beta} + 3\tilde{\alpha}_{Cl} \tilde{\beta} - 6\tilde{\beta}^2 - 12\tilde{\beta}_{Cl}^2 - (2\tilde{\alpha} + \tilde{\alpha}_{Cl} + \tilde{\beta})^2 \right) \sqrt[3]{-27\tilde{\alpha}_s^2 \alpha_{s_{Cl}}}}{8 \sqrt[3]{2} (1-\sqrt{3}i)(1+\sqrt{3}i)}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $?_1 Norm^4 \tilde{\delta} + ?_2 Norm^4 \tilde{\delta} + ?_3 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 1, 1, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 1, 0, 0, 2, 2), (0, 0, 0, 0, 1, 0, 0, 0)$

- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$

- η_5 : linear combination = 0, energy = $\alpha_{s_{Cl}}$

- Δ Energy: $-\tilde{\alpha}_{Cl} + \alpha_{s_{Cl}}$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i) \left(88\tilde{\alpha} + 32\tilde{\alpha}_{Cl} - 4\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2 \tilde{\alpha}_{Cl} - 27\tilde{\alpha} \tilde{\alpha}_{Cl} \tilde{\beta} + 108\tilde{\alpha} \tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl} \tilde{\beta}^2} \right)}{(1-\sqrt{3}i)(1+\sqrt{3}i)}$

- Energy of State after Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i) \left(88\tilde{\alpha} + 20\tilde{\alpha}_{Cl} + 12\alpha_{s_{Cl}} - 4\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2 \tilde{\alpha}_{Cl} - 27\tilde{\alpha} \tilde{\alpha}_{Cl} \tilde{\beta} + 108\tilde{\alpha} \tilde{\beta}_{Cl}^2} \right)}{(1-\sqrt{3}i)(1+\sqrt{3}i)}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 0, 1, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation: $(1-\sqrt{3}i) \left(46\tilde{\alpha} + 14\tilde{\alpha}_{Cl} + 2\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)$

- Energy of State after Excitation: $(1-\sqrt{3}i) \left(40\tilde{\alpha} + 14\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s - 4\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation: $(1-\sqrt{3}i) \left(46\tilde{\alpha} + 14\tilde{\alpha}_{Cl} + 2\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)$

- Energy of State after Excitation: $(1-\sqrt{3}i) \left(40\tilde{\alpha} + 14\tilde{\alpha}_{Cl} + 6\tilde{\alpha}_s - 4\tilde{\beta} + 6\tilde{\beta}_s - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 2, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2, 2), (0, 0, 0, 0, 1, 0, 0, 0)$

- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$

- η_5 : linear combination = 0, energy = $\alpha_{s_{Cl}}$

- Δ Energy: $-\tilde{\alpha}_{Cl} + \alpha_{s_{Cl}}$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 14\tilde{\alpha}_{Cl} + 2\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{1}$$

- Energy of State after Excitation:
$$\frac{(1-\sqrt{3}i) \left(46\tilde{\alpha} + 8\tilde{\alpha}_{Cl} + 6\alpha_{s_{Cl}} + 2\tilde{\beta} - 2 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2} \right)}{1}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 1, 0, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$

- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$

- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$

- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i)(1+\sqrt{3}i) \left(100\tilde{\alpha} + 20\tilde{\alpha}_{Cl} + 8\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2} \right)}{1}$$

- Energy of State after Excitation:
$$\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i) \left(3\tilde{\alpha}^2 + 6\tilde{\alpha}\tilde{\alpha}_{Cl} + 3\tilde{\alpha}\tilde{\beta} + 3\tilde{\alpha}_{Cl}\tilde{\beta} - 6\tilde{\beta}^2 - 12\tilde{\beta}_{Cl}^2 - (2\tilde{\alpha} + \tilde{\alpha}_{Cl} + \tilde{\beta})^2 \right) \left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 108\tilde{\beta}\tilde{\beta}_{Cl}^2 \right)}{1}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right)}$$

- Energy of State after Excitation:
$$\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right)}$$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 2, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2, 2), (0, 0, 0, 1, 0, 0, 0, 0)$

- ψ_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right)$, energy =
$$\frac{(1+\sqrt{3}i)\left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}(1+\sqrt{3}i)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}\right)}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta})^2)}}+(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl})}}$$

- η_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right)$, energy =
$$\frac{(1+\sqrt{3}i)\left(8\tilde{\alpha}_s+4\alpha_{s_{Cl}}+4\tilde{\beta}_s+2^{2/3}(1+\sqrt{3}i)\sqrt[3]{-27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl}}\right)}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}_s\tilde{\alpha}_{Cl}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s\tilde{\beta}_{s_{Cl}}^2+27\alpha_{s_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s\tilde{\beta}_{s_{Cl}}^2+\sqrt{-4(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s\tilde{\alpha}_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3\alpha_{s_{Cl}}\tilde{\beta}_s+6\tilde{\beta}_s^2+12\tilde{\beta}_{s_{Cl}}^2+(-2\tilde{\alpha}_s-\tilde{\alpha}_{Cl}-\tilde{\beta}_s)^2)}}+(-27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl})}}$$

- Δ Energy:
$$-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{\tilde{\beta}}{3} + \frac{\tilde{\beta}_s}{3} + \frac{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta})^2)}}+(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl})}}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}_s\tilde{\alpha}_{Cl}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s\tilde{\beta}_{s_{Cl}}^2+27\alpha_{s_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s\tilde{\beta}_{s_{Cl}}^2+\sqrt{-4(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s\tilde{\alpha}_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3\alpha_{s_{Cl}}\tilde{\beta}_s+6\tilde{\beta}_s^2+12\tilde{\beta}_{s_{Cl}}^2+(-2\tilde{\alpha}_s-\tilde{\alpha}_{Cl}-\tilde{\beta}_s)^2)}}+(-27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl})}}$$

$$3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2\alpha_{s_{Cl}}}{2}-\frac{27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s\beta_{s_{Cl}}^2+27\alpha_{s_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s\beta_{s_{Cl}}^2+\sqrt{-4(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s\alpha_{s_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3\alpha_{s_{Cl}}\tilde{\beta}_s+6\tilde{\beta}_s^2+12\beta_{s_{Cl}}^2+(-2\tilde{\alpha}_s-\alpha_{s_{Cl}}-\tilde{\beta}_s)^2)}}+(-27\tilde{\alpha}_s^2\alpha_{s_{Cl}})}}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta})^2)}}+(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl})}}$$

$$\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2\alpha_{s_{Cl}}}{2}-\frac{27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s\beta_{s_{Cl}}^2+27\alpha_{s_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s\beta_{s_{Cl}}^2+\sqrt{-4(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s\alpha_{s_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3\alpha_{s_{Cl}}\tilde{\beta}_s+6\tilde{\beta}_s^2+12\beta_{s_{Cl}}^2+(-2\tilde{\alpha}_s-\alpha_{s_{Cl}}-\tilde{\beta}_s)^2)}}+(-27\tilde{\alpha}_s^2\alpha_{s_{Cl}})}}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta})^2)}}+(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl})}}$$

- Energy of State before Excitation:
$$\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}\right)}{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right)}$$

$$8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2)\sqrt[3]{-27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl}}$$

- Energy of State after Excitation: _____

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

$$(1-\sqrt{3}i)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)$$

- Energy of State before Excitation: _____

$$(1-\sqrt{3}i)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)$$

- Energy of State after Excitation: _____

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 0, 1, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

$$(1-\sqrt{3}i)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)$$

- Energy of State before Excitation: _____

$$(1-\sqrt{3}i)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)$$

- Energy of State after Excitation: _____

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 2, 1, 0, 1, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2, 2), (0, 0, 0, 0, 1, 0, 0, 0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$
- η_5 : linear combination = 0, energy = $\alpha_{s_{Cl}}$
- Δ Energy: $-\tilde{\alpha}_{Cl} + \alpha_{s_{Cl}}$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)}{(1-\sqrt{3}i)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6\alpha_{s_{Cl}}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)}$

- Energy of State after Excitation: $\frac{(1-\sqrt{3}i)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6\alpha_{s_{Cl}}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)}{(1-\sqrt{3}i)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6\alpha_{s_{Cl}}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\sqrt{3}(\mathbf{p}_1 - \mathbf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}(\mathbf{s}_1 - \mathbf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right)}{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2\right)}$

- Energy of State after Excitation: $\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2\right)}{(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2\right)}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s3} :

- $-(2, 2, 1, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 1, 0, 0, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)}{8} \left(100\tilde{\alpha} + 20\tilde{\alpha}_{Cl} + 8\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2} \right)$

- Energy of State after Excitation: $\frac{8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)}{3} \left(3\tilde{\alpha}^2 + 6\tilde{\alpha}\tilde{\alpha}_{Cl} + 3\tilde{\alpha}\tilde{\beta} + 3\tilde{\alpha}_{Cl}\tilde{\beta} - 6\tilde{\beta}^2 - 12\tilde{\beta}_{Cl}^2 - (2\tilde{\alpha} + \tilde{\alpha}_{Cl} + \tilde{\beta})^2 \right) \left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} \right)$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s4} :

- $-(2, 2, 1, 1, 0, 0, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2, 2), (0, 0, 0, 1, 0, 0, 0, 0)$

- ψ_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}p_1}{2} + \frac{\sqrt{2}p_4}{2} \right)$, energy = $\frac{(1+\sqrt{3}i)}{3} \left(8\tilde{\alpha} + 4\tilde{\alpha}_{Cl} + 4\tilde{\beta} + 2^{2/3} (1+\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}} \right)$

- η_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}s_1}{2} + \frac{\sqrt{2}s_4}{2} \right)$, energy = $\frac{(1+\sqrt{3}i)}{3} \left(8\tilde{\alpha}_s + 4\alpha_{s_{Cl}} + 4\tilde{\beta}_s + 2^{2/3} (1+\sqrt{3}i) \sqrt[3]{-27\tilde{\alpha}_s^2\tilde{\alpha}_{Cl}} \right)$

- Δ Energy: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{\tilde{\beta}}{3} + \frac{\tilde{\beta}_s}{3} + \frac{3\left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) \sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2} - \frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2} + 54\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 27\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 54\tilde{\beta}\tilde{\beta}_{Cl}^2 + \sqrt{-4(-3\tilde{\alpha}^2 - 6\tilde{\alpha}\tilde{\alpha}_{Cl} - 3\tilde{\alpha}\tilde{\beta} - 3\tilde{\alpha}_{Cl}\tilde{\beta} + 6\tilde{\beta}^2 + 12\tilde{\beta}_{Cl}^2 + (-2\tilde{\alpha} - \tilde{\alpha}_{Cl} - \tilde{\beta})^2)}}}{3}$

$$\frac{3\left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) \sqrt[3]{-\frac{27\tilde{\alpha}_s^2\alpha_{s_{Cl}}}{2} - \frac{27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s}{2} + 54\tilde{\alpha}_s\beta_{s_{Cl}}^2 + 27\alpha_{s_{Cl}}\tilde{\beta}_s^2 + 54\tilde{\beta}_s\beta_{s_{Cl}}^2 + \sqrt{-4(-3\tilde{\alpha}_s^2 - 6\tilde{\alpha}_s\alpha_{s_{Cl}} - 3\tilde{\alpha}_s\tilde{\beta}_s - 3\alpha_{s_{Cl}}\tilde{\beta}_s + 6\tilde{\beta}_s^2 + 12\beta_{s_{Cl}}^2 + (-2\tilde{\alpha}_s - \alpha_{s_{Cl}} - \tilde{\beta}_s)^2)}}}{3}$$

$$\frac{\left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) \sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2} - \frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2} + 54\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 27\tilde{\alpha}_{Cl}\tilde{\beta}^2 + 54\tilde{\beta}\tilde{\beta}_{Cl}^2 + \sqrt{-4(-3\tilde{\alpha}^2 - 6\tilde{\alpha}\tilde{\alpha}_{Cl} - 3\tilde{\alpha}\tilde{\beta} - 3\tilde{\alpha}_{Cl}\tilde{\beta} + 6\tilde{\beta}^2 + 12\tilde{\beta}_{Cl}^2 + (-2\tilde{\alpha} - \tilde{\alpha}_{Cl} - \tilde{\beta})^2)}}}{3} + \frac{\left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) \sqrt[3]{-\frac{27\tilde{\alpha}_s^2\alpha_{s_{Cl}}}{2} - \frac{27\tilde{\alpha}_s\alpha_{s_{Cl}}\tilde{\beta}_s}{2} + 54\tilde{\alpha}_s\beta_{s_{Cl}}^2 + 27\alpha_{s_{Cl}}\tilde{\beta}_s^2 + 54\tilde{\beta}_s\beta_{s_{Cl}}^2 + \sqrt{-4(-3\tilde{\alpha}_s^2 - 6\tilde{\alpha}_s\alpha_{s_{Cl}} - 3\tilde{\alpha}_s\tilde{\beta}_s - 3\alpha_{s_{Cl}}\tilde{\beta}_s + 6\tilde{\beta}_s^2 + 12\beta_{s_{Cl}}^2 + (-2\tilde{\alpha}_s - \alpha_{s_{Cl}} - \tilde{\beta}_s)^2)}}}{3}$$

- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)(1+\sqrt{3}i)}{8} \left(100\tilde{\alpha} + 20\tilde{\alpha}_{Cl} + 8\tilde{\beta} - 4 \cdot 2^{2/3} \sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl} - 27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta} + 108\tilde{\alpha}\tilde{\beta}_{Cl}^2 + 54\tilde{\alpha}_{Cl}\tilde{\beta}^2} \right)$

$$8\sqrt[3]{2}(1-\sqrt{3}i)(1+\sqrt{3}i)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta})^2\right)\sqrt[3]{-27\tilde{\alpha}_s^2a}$$

- Energy of State after Excitation: _____

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{\alpha_{s_{Cl}}}{4}$

- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta}$

- AND

Dispersion energies following from the transitions

1. State: A = [2, 2, 2, 0, 0, 0, 2, 2] $\leftrightarrow$ B = (2, 1, 1, 1, 1, 0, 2, 2)

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{32\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} \right)}$$

2. State: A = (2, 1, 1, 1, 1, 0, 2, 2) $\leftrightarrow$ B = [2, 2, 2, 0, 0, 0, 2, 2]

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{32\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} \right)}$$

3. State: A = (2, 1, 2, 0, 1, 0, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4 \left(-12 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) - 9 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}}) \right)}{r^6 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}})}$$

4. State: A = (2, 1, 2, 0, 1, 0, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{Cl}}{3} \right)}$$

5. State: A = (2, 1, 2, 0, 1, 0, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4 \left(-12 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) - 9 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}}) \right)}{r^6 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}})}$$

6. State: A = (2, 1, 2, 0, 1, 0, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{Cl}}{3} \right)}$$

7. State: $A = (2, 1, 2, 1, 0, 0, 2, 2) \leftrightarrow B = (2, 1, 2, 0, 1, 0, 2, 2)$

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_C}{3} \right)}$$

8. State: $A = (2, 1, 2, 1, 0, 0, 2, 2) \leftrightarrow B = (2, 1, 2, 1, 0, 0, 2, 2)$

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{1}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)^3 \sqrt{-\frac{1}{2} - \frac{\sqrt{3}i}{2}}} \right)}$$

9. State: $A = (2, 1, 2, 1, 0, 0, 2, 2) \leftrightarrow B = (2, 2, 1, 0, 1, 0, 2, 2)$

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_C}{3} \right)}$$

10. State: $A = (2, 1, 2, 1, 0, 0, 2, 2) \leftrightarrow B = (2, 2, 1, 1, 0, 0, 2, 2)$

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{1}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)^3 \sqrt{-\frac{1}{2} - \frac{\sqrt{3}i}{2}}} \right)}$$

11. State: $A = (2, 2, 1, 0, 1, 0, 2, 2) \leftrightarrow B = (2, 1, 2, 0, 1, 0, 2, 2)$

$$E_{dispersion} = \frac{2\tilde{\delta}^4 \left(-12(\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right) - 9(\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) \left(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}} \right) \right)}{r^6 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right) \left(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}} \right)}$$

12. State: $A = (2, 2, 1, 0, 1, 0, 2, 2) \leftrightarrow B = (2, 1, 2, 1, 0, 0, 2, 2)$

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{1}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_C}{3} \right)}$$

13. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = -\frac{2\tilde{\delta}^4 \left(-12 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) - 9 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}}) \right)}{r^6 (\tilde{\alpha}_{Cl} - \alpha_{s_{Cl}}) (\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s) (\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - \alpha_{s_{Cl}})}$$

14. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{Cl}}{3} \right)}$$

15. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{Cl}}{3} \right)}$$

16. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{36\tilde{\delta}^4}{3 \left(-\frac{1}{2} - \frac{\sqrt{3}i}{2} \right)^3 \sqrt{-\frac{1}{2} - \frac{\sqrt{3}i}{2}}} \right)}$$

17. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6 \left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + \alpha_{s_{Cl}} - \tilde{\beta} + \tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{Cl}}{3} \right)}$$

18. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{36\tilde{\delta}^4}{r^6 \left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{\alpha_{s_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{36\tilde{\delta}^4}{3 \left(-\frac{1}{2} - \frac{\sqrt{3}i}{2} \right)^3 \sqrt{-\frac{1}{2} - \frac{\sqrt{3}i}{2}}} \right)}$$

Ground-State:

A = [2, 2, 2, 0, 0, 0, 2, 2] ↔ B = [2, 2, 2, 0, 0, 0, 2, 2]

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$